MTH2021 Mathematical Methods 2

Self-study sheet 2 – solutions

1.
(a) $f(z) = \operatorname{Im} z = y$. At a given z, $f(z)$ varies with y, i.e. along the direction $x = \text{const}$, but is constant along $y = \text{const}$. Thus it is not differentiable (in the sense of differentiability in complex analysis) at any z. Hence, it is nowhere analytic.

(b) Now it is easiest to write z in polar form, $z = r\exp(\mathrm{i}\theta)$. Then

$$f(z) = \frac{z^2}{z^{*2}} = \frac{\exp(2\mathrm{i}\theta)}{\exp(-2\mathrm{i}\theta)} = \exp(4\mathrm{i}\theta) \tag{1}$$

At a given z, $f(z)$ is constant along the radial direction, i.e. along the ray $\theta = \text{const}$, but varies with θ, i.e. along the circle $r = \text{const}$. Thus, it is nowhere analytic. It is, further, undefined at $z = 0$.

2.
(a) We have

$$z^4 = (x + \mathrm{i}y)^4 = (x^4 - 6x^2y^2 + y^4) + \mathrm{i}(4x^3y - 4xy^3) = u + \mathrm{i}v \tag{2}$$

where

$$u = x^4 - 6x^2y^2 + y^4 \qquad v = 4x^3y - 4xy^3 \tag{3}$$

The partial derivatives are

$$u_x = 4x^3 - 12xy^2 \qquad u_y = -12x^2y + 4y^3 \tag{4}$$
$$v_x = 12x^2y - 4y^3 \qquad v_y = 4x^3 - 12xy^2 \tag{5}$$

They are continuous and satisfy the Cauchy-Riemann equations ($u_x = v_y$, $u_y = -v_x$) everywhere. The function is analytic everywhere.

(b) We have

$$z^2 - (z^*)^2 = \mathrm{i}\,4xy = u + \mathrm{i}v \tag{6}$$

where

$$u = 0 \qquad v = 4xy \tag{7}$$

The partial derivatives are

$$u_x = 0 \qquad u_y = 0 \tag{8}$$
$$v_x = 4y \qquad v_y = 4x \tag{9}$$

The Cauchy-Riemann equations are satisfied only at $z = 0$. An analytic function must be differentiable at a point and throughout a neighbourhood of that point. The function is not analytic anywhere.

(c) We have

$$\operatorname{Im} z = y = u + \mathrm{i}v \tag{10}$$

where

$$u = y \qquad v = 0 \tag{11}$$

The partial derivatives are

$$u_x = 0 \qquad u_y = 1 \tag{12}$$
$$v_x = 0 \qquad v_y = 0 \tag{13}$$

The Cauchy-Riemann equations are not satisfied anywhere. The function is not analytic anywhere.

(d) We have

$$\frac{1}{z+\mathrm{i}} = \frac{x - \mathrm{i}(y+1)}{x^2 + (y+1)^2} = u + \mathrm{i}v \tag{14}$$

where

$$u = \frac{x}{x^2 + (y+1)^2} \qquad v = -\frac{y+1}{x^2 + (y+1)^2} \tag{15}$$

The partial derivatives are

$$u_x = \frac{1}{x^2 + (y+1)^2} - \frac{2x^2}{(x^2 + (y+1)^2)^2} \qquad u_y = -\frac{2x(y+1)}{(x^2 + (y+1)^2)^2} \tag{16}$$
$$v_x = \frac{2x(y+1)}{(x^2 + (y+1)^2)^2} \qquad v_y = \frac{2(y+1)^2}{(x^2 + (y+1)^2)^2} - \frac{1}{x^2 + (y+1)^2} \tag{17}$$

The Cauchy-Riemann equations are satisfied at all $z \neq -\mathrm{i}$. At $z = -\mathrm{i}$ the function is undefined. The function is analytic everywhere except at $z = -\mathrm{i}$.

(e) We have

$$\cos z = \cos x \cosh y - \mathrm{i} \, \sin x \sinh y = u + \mathrm{i}v \tag{18}$$

where

$$u = \cos x \cosh y \qquad v = -\sin x \sinh y \tag{19}$$

The partial derivatives are

$$u_x = -\sin x \cosh y \qquad u_y = \cos x \sinh y \tag{20}$$
$$v_x = -\cos x \sinh y \qquad v_y = -\sin x \cosh y \tag{21}$$

The Cauchy-Riemann equations are satisfied everywhere. The function is analytic everywhere.

(f) We have

$$\sinh z = \sinh x \cos y + \mathrm{i} \, \cosh x \sin y = u + \mathrm{i}v \tag{22}$$

where

$$u = \sinh x \cos y \qquad v = \cosh x \sin y \tag{23}$$

The partial derivatives are

$$u_x = \cosh x \cos y \qquad u_y = -\sinh x \sin y \tag{24}$$

$$v_x = \sinh x \sin y \qquad v_y = \cosh x \cos y \tag{25}$$

The Cauchy-Riemann equations are satisfied everywhere. The function is analytic everywhere.

3.
First:

$$\begin{aligned} f(z+\Delta z) - f(z) &= (z+\Delta z)^n - z^n & (26)\\ &= z^n + nz^{n-1}\Delta z + \frac{n(n-1)}{2} z^{n-2}(\Delta z)^2 + ... + (\Delta z)^n - z^n & (27)\\ &= nz^{n-1}\Delta z + \frac{n(n-1)}{2} z^{n-2}(\Delta z)^2 + ... + (\Delta z)^n & (28)\end{aligned}$$

Next:

$$\frac{f(z+\Delta z) - f(z)}{\Delta z} = nz^{n-1} + \frac{n(n-1)}{2} z^{n-2}\Delta z + ... + (\Delta z)^{n-1} \tag{29}$$

Finally we must consider the limit of this quantity as $\Delta z \to 0$. According to a theorem on limits, the limit of a sum of a finite number of terms exists, and is equal to the sum of limits, provided the limits of terms exist individually. This is the case above and so

$$\begin{aligned} f'(z) = \lim_{\Delta z \to 0} \frac{f(z+\Delta z) - f(z)}{\Delta z} &= \lim_{\Delta z \to 0} \left[nz^{n-1} + \frac{n(n-1)}{2} z^{n-2}\Delta z + ... + (\Delta z)^{n-1} \right] & (30)\\ &= nz^{n-1} & (31)\end{aligned}$$

4.
We can write the definitions

$$\cos z = \frac{\exp(\mathrm{i}z) + \exp(-\mathrm{i}z)}{2} \tag{32}$$

$$\sin z = \frac{\exp(\mathrm{i}z) - \exp(-\mathrm{i}z)}{2\mathrm{i}} \tag{33}$$

as

$$2\cos z = \exp(\mathrm{i}z) + \exp(-\mathrm{i}z) \tag{34}$$

$$2\mathrm{i}\,\sin z = \exp(\mathrm{i}z) - \exp(-\mathrm{i}z) \tag{35}$$

Adding and subtracting the last two equations:

$$\exp(\pm \mathrm{i}z) = \cos z \pm \mathrm{i} \sin z \tag{36}$$

In an analogous way, we can write

$$\cosh z = \frac{\exp z + \exp(-z)}{2} \tag{37}$$

$$\sinh z = \frac{\exp z - \exp(-z)}{2} \tag{38}$$

as

$$\exp(\pm z) = \cosh z \pm \sinh z \tag{39}$$

(a)

$$\begin{aligned}
\cos(z_1 + z_2) &= \frac{1}{2}\left[\exp(\mathrm{i}(z_1 + z_2)) + \exp(-\mathrm{i}(z_1 + z_2))\right] && (40)\\
&= \frac{1}{2}\left[\exp(\mathrm{i}z_1)\exp(\mathrm{i}z_2) + \exp(-\mathrm{i}z_1)\exp(-\mathrm{i}z_2))\right] && (41)\\
&= \frac{1}{2}\left[(\cos z_1 + \mathrm{i}\sin z_1)(\cos z_2 + \mathrm{i}\sin z_2)\right.\\
&\quad \left.+(\cos z_1 - \mathrm{i}\sin z_1)(\cos z_2 - \mathrm{i}\sin z_2)\right] && (42)\\
&= \cos z_1 \cos z_2 - \sin z_1 \sin z_2 && (43)
\end{aligned}$$

(b)

$$\cos(\mathrm{i}z) = \frac{\exp(\mathrm{i}(\mathrm{i}z)) + \exp(-\mathrm{i}(\mathrm{i}z))}{2} = \frac{\exp(-z) + \exp z}{2} = \cosh z \tag{44}$$

(c)

$$\begin{aligned}
\sinh(z_1 + z_2) &= \frac{1}{2}\left[\exp(z_1 + z_2) - \exp(-(z_1 + z_2))\right] && (45)\\
&= \frac{1}{2}\left[\exp(z_1)\exp(z_2) - \exp(-z_1)\exp(-z_2)\right] && (46)\\
&= \frac{1}{2}\left[(\cosh z_1 + \sinh z_1)(\cosh z_2 + \sinh z_2)\right.\\
&\quad \left.-(\cosh z_1 - \sinh z_1)(\cosh z_2 - \sinh z_2)\right] && (47)\\
&= \sinh z_1 \cosh z_2 + \cosh z_1 \sinh z_2 && (48)
\end{aligned}$$

(d)

$$\sinh(\mathrm{i}z) = \frac{\exp(\mathrm{i}z) - \exp(-\mathrm{i}z)}{2} = \mathrm{i}\sin z \tag{49}$$

Put $z = \pi$ in (d): $\sinh(\mathrm{i}\pi) = \mathrm{i}\sin\pi = 0$. Put $z = \mathrm{i}\pi/3$ in (b): $\cosh(\mathrm{i}\pi/3) = \cos(-\pi/3) = \cos(\pi/3) = 1/2$.

5.
(a)

$$\log(-2) = \log|-2| + \mathrm{i}\,\arg(-2) = \log 2 + \mathrm{i}(\pi + 2k\pi) \quad , \quad k = 0, \pm 1, \pm 2, \ldots \tag{50}$$

(b)

$$\begin{aligned}
\exp(\mathrm{i}z) = 1 + \mathrm{i} \quad \Rightarrow \quad \mathrm{i}z &= \log(1 + \mathrm{i}) && (51)\\
&= \log(\sqrt{2}) + \mathrm{i}(\pi/4 + 2k\pi) && (52)\\
&\Rightarrow \quad z = -\frac{\mathrm{i}}{2}\log 2 + \pi/4 + 2k\pi \quad , \quad k = 0, \pm 1, \pm 2, \ldots && (53)
\end{aligned}$$